Transformations: Mini-Lessons

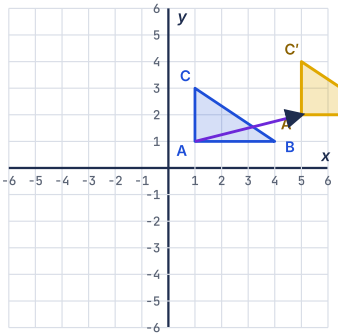
8.1 Translations (slides) NC.8.G.3

A **translation** slides a figure. **Right/left changes x** (add or subtract). **Up/down changes y**. The image is **congruent** (same size and shape).

Pre-image
the original figure

Image
the figure after the move, named with ′ (A′, "A prime")

Congruent
same size and same shape



A(1, 1) → A′(5, 2)
B(4, 1) → B′(8, 2)
C(1, 3) → C′(5, 4)

(x, y) → (x + 4, y + 1)

same size and shape: congruent

Move	right h	left h	up k	down k
Do	x + h	x − h	y + k	y − k

Watch out: Left and right change **x** (the first number). Up and down change **y** (the second number).

Walk through: `Translate (1, -2) left 3 and up 5.`

1. Left 3 changes which number?
   x: 1 − 3 = −2 → x: 1 − 3 = −2
2. Up 5 changes which number?
   y: −2 + 5 = 3 → y: −2 + 5 = 3
3. The image point is…
   (−2, 3) → (1, −2) → (−2, 3)

Check your understanding

1 Score 3 NC.8.G.3

Translate right 4 and up 1: the point **(2, 3)**. Where does it land?

Ⓐ (−2, 4) Ⓑ (3, 7)

Ⓒ **(6, 4)** Ⓓ (6, 2)

2 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

Ⓐ (0, 8) Ⓑ (−1, 3)

Ⓒ (4, 8) Ⓓ **(0, 2)**

8.2 Reflections (flips) NC.8.G.3

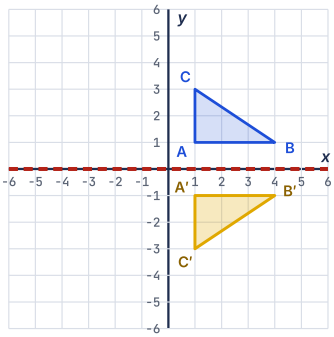
A **reflection** flips a figure over a line. Over the **x-axis**: y changes sign, `(x, y) → (x, -y)`. Over the **y-axis**: x changes sign, `(x, y) → (-x, y)`.

Line of reflection

the mirror line

Opposite

same number, other sign: the opposite of 5 is −5



`A(1, 1) → A'(1, -1)`
`B(4, 1) → B'(4, -1)`
`C(1, 3) → C'(1, -3)`

`(x, y) → (x, -y)`

over the x-axis: y changes sign

Trick: you flip **over** the x-axis, so the **other** letter (y) changes sign.

Watch out: Only **one** coordinate changes sign in a reflection over an axis.

Walk through: `Reflect (3, 5) over the y-axis.`

1 Over the y-axis, which coordinate changes sign?
x → x changes sign

2 The image is...
(−3, 5) → (3, 5) → (−3, 5)

Check your understanding

1 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(3, 5)**. Where does it land?

Ⓐ **(3, −5)** Ⓑ (−3, 5)

Ⓒ (5, 3) Ⓓ (−3, −5)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

Ⓐ **(3, 4)** Ⓑ (4, −3)

Ⓒ (3, −4) Ⓓ (−3, −4)

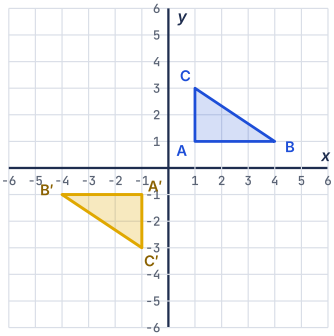
8.3 Rotations (turns) NC.8.G.3

A **rotation** turns a figure around the origin. **180°**: `(x, y) → (-x, -y)`. **90° counterclockwise**: `(x, y) → (-y, x)`. **270° counterclockwise** (90° clockwise): `(x, y) → (y, -x)`.

Counterclockwise
the opposite way a clock's hands turn

Origin
the point (0, 0) where the axes cross

Turn	Rule	(4, 2) →
90° CCW	(−y, x)	(−2, 4)
180°	(−x, −y)	(−4, −2)
270° CCW	(y, −x)	(2, −4)



`A(1, 1) → A'(-1, -1)`
`B(4, 1) → B'(-4, -1)`
`C(1, 3) → C'(-1, -3)`

`(x, y) → (-x, -y)`
180° turn: both signs change

Watch out: For 90° and 270°, the numbers **switch places**. For 180°, they do not.

Walk through: `Rotate (1, 5) 90° counterclockwise.`

1. Which rule is 90° counterclockwise?
   (x, y) → (−y, x) → (x, y) → (−y, x)
2. Here x = 1 and y = 5. What is −y?
   −5 → −y = −5
3. The image is…
   (−5, 1) → (1, 5) → (−5, 1)

Check your understanding

1 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(4, 2)**. Where does it land?

Ⓐ (2, 4) Ⓑ **(−4, −2)**
Ⓒ (−4, 2) Ⓓ (−2, 4)

2 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

Ⓐ (5, 5) Ⓑ (−5, −5)
Ⓒ (5, −5) Ⓓ **(−5, 5)**

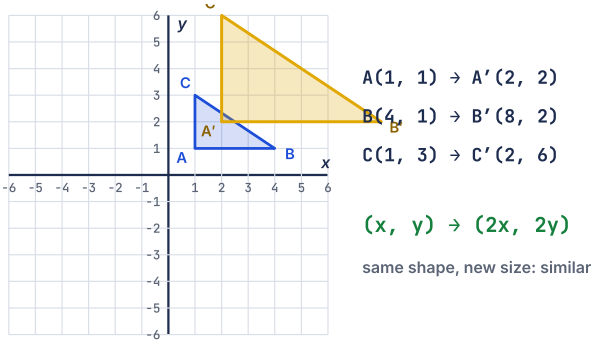
8.4 Dilations, congruent and similar NC.8.G.4

A **dilation** multiplies both coordinates by the **scale factor k**: `(x, y) → (kx, ky)`. k > 1 makes it bigger; k between 0 and 1 makes it smaller. The image is **similar**. Slides, flips and turns keep the image **congruent**.

Scale factor (k)
the number you multiply by

Similar
same shape, maybe a different size

Congruent
same shape AND same size



Dilation = resize = similar. Translation, reflection, rotation = same size = congruent.

Watch out: Multiply **both** numbers by k. Do not add k.

Walk through: `Dilate (3, -4) by k = 3.`

1. What do we do with k = 3?
   Multiply both coordinates by 3 → (3 · 3, −4 · 3)
2. Multiply.
   (9, −12) → (3, −4) → (9, −12)
3. Is the image congruent or similar?
   Similar: same shape, bigger size → similar (not congruent)

Check your understanding

1 Score 3 NC.8.G.4

Dilate by k = 2: the point **(2, 3)**. Where does it land?

- **(A) (4, 6)**
- (B) (4, 5)
- (C) (1, 1.5)
- (D) (2, 6)

2 Score 3 NC.8.G.4

After a **dilation** with k = 3, the image is...

- **(A) similar to the pre-image**
- (B) congruent to the pre-image
- (C) neither congruent nor similar

3 Score 3 NC.8.G.2

After a **reflection**, the image is...

A congruent to the pre-image

B bigger than the pre-image

C smaller than the pre-image